

Using LALMs for Pronunciation Error Detection in Arabic

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Abstract

This paper presents Qwen-MDD-AR, our system for the Iqra’Eval.2 Interspeech 2026 challenge on Modern Standard Arabic pronunciation assessment. Built on Qwen3-ASR-1.7B, it frames Arabic mispronunciation detection and diagnosis as autoregressive speech-to-phoneme generation. We add the 68-phoneme MSA inventory as special tokens, freeze the audio encoder, and adapt the decoder with LoRA. Using only provided challenge data, our submission ranked 6th of 19 teams on the hidden QuranMB.v2 test set, with 0.6702 F1 and 0.0445 PER. In a controlled comparison on our internal mixed held-out set, a Wav2Vec2-BERT CTC baseline trained on the same data achieved better PER, precision, and F1, while Qwen-MDD-AR obtained the same recall and a small Correct Diagnosis advantage (0.9496 vs. 0.9328). Results suggest LoRA-adapted LALMs are a practical alternative for Arabic MDD, while alignment accuracy and inference cost remain open.

1 Introduction

Automatic pronunciation assessment is central to Computer-Assisted Pronunciation Training (CAPT), where learners need scalable and localized feedback on spoken errors. Mispronunciation Detection and Diagnosis (MDD) remains difficult for morphologically rich and phonetically complex languages such as Modern Standard Arabic (MSA), where small phonetic confusions can change the pedagogical feedback required (Kheir et al., 2023).

Recent Arabic MDD work has benefited from standardized benchmarks and phoneme inventories, including QuranMB.v1 and the Iqra’Eval shared tasks (Kheir et al., 2025; El Kheir et al., 2025). Iqra’Eval 2 adds Iqra_Extra_IS26, a roughly two-hour set of real human mispronunciations, making it a useful setting for testing whether models trained mostly on clean or synthetic data can diagnose real errors.

Most high-performing Arabic MDD systems use CTC models built on self-supervised speech representations such as Wav2Vec2 (Graves et al., 2006; Baevski et al., 2020). Strong Iqra’Eval systems such as AraS2P and Hafs2Vec used multi-stage adaptation and Tajweed-aware phonemizers, often with external acoustic data (Mattar et al., 2025; Ibrahim, 2025). In parallel, autoregressive speech models and Large Audio-Language Models (LALMs) have become plausible alternatives for speech-to-phoneme generation, although they do not remove inference-cost concerns and are not uniformly better than CTC systems.

The motivation for using a LALM is therefore not to replace CTC systems outright, but to test a different inductive bias under the same challenge data constraints. CTC models are naturally suited to monotonic acoustic-to-token alignment, while autoregressive decoders can condition each generated phoneme on a broader output history and the model’s pretrained speech-language representations. For a small real-error dataset, this broader context may help diagnosis, but it can also introduce risks: the decoder may rely on linguistic priors, smooth over acoustically present errors, or generate extra tokens. A careful comparison must therefore report both the strengths and weaknesses of the generative formulation.

We present Qwen-MDD-AR, a LoRA-adapted Qwen3-ASR-1.7B system for autoregressive speech-to-phoneme generation with 68 atomic MSA phoneme tokens (Shi et al., 2026; Hu et al., 2021). Unlike prior Iqra’Eval systems centered on acoustic-model adaptation and specialized phonemization (Mattar et al., 2025; Ibrahim, 2025), our contribution is an empirical investigation of this recent LALM family for Arabic MDD, not a new architecture or fine-tuning algorithm. We evaluate it under challenge-data constraints against a matched-data Wav2Vec2-BERT CTC baseline (Communication et al., 2023) and analyze recognition errors and

diagnosis outcomes; these results do not establish architectural or diagnostic superiority.

2 Related Work

Self-supervised acoustic models with CTC remain strong for Arabic MDD. Prior Iqra’Eval systems adapted Wav2Vec2 and HuBERT variants, while AraS2P and Hafs2Vec showed the value of domain-specific pre-training and phonemization (El Kheir et al., 2025; Baevski et al., 2020; Hsu et al., 2021; Mattar et al., 2025; Ibrahim, 2025). Transformer-based Arabic MDD systems have also been used to detect insertion, deletion, and substitution errors (Alrashoudi et al., 2025). Our work keeps this CTC line as a baseline and studies a different adaptation path.

Autoregressive speech models provide another route by generating phoneme sequences directly. Arabic evidence includes MUBEEN, which adapted a Whisper encoder for pronunciation assessment and showed that encoder-decoder models can be competitive in a shared-task setting (Radford et al., 2022; El Kheir et al., 2025). Related English work reports gains from instruction-tuned ALMs over non-fine-tuned or cascaded baselines for mispronunciation detection and feedback generation (Liu et al., 2026; Wu et al., 2024). We treat these studies as motivation rather than direct evidence that LALMs dominate MDD, since their datasets, languages, and baselines differ from the present Arabic challenge.

Parameter-efficient fine-tuning is especially relevant when adapting billion-parameter speech models to small task-specific datasets. LoRA reduces the number of trainable parameters by injecting low-rank updates into selected projection matrices (Hu et al., 2021), and has been used in speech settings such as multi-accent ASR (Bagat et al., 2025). Ahn and Nam (Ahn and Nam, 2025) show that a LoRA-tuned multimodal LLM can support a simpler APA/MDD methodology, while also indicating that specialized systems can retain advantages on core PER metrics. We adapt this PEFT direction to Arabic MDD and compare it with a Wav2Vec2-BERT CTC baseline trained on identical data.

3 System Description

3.1 Task and Data

Qwen-MDD-AR frames MDD as speech-to-phoneme generation. The Iqra’Eval output is a whitespace-separated phoneme sequence, e.g.,

l a < a k a l u u m i n f a w q I h i m. We added the 68 MSA phonemes as atomic special tokens so the model can generate the target phoneme inventory directly. The inventory follows the MSA phonetizer of Halabi and Wald (Halabi and Wald, 2016), which represents gemination by doubling the consonant symbol (e.g., b → bb); it comprises 40 base symbols (12 vowels and 28 consonants) plus 28 geminates, and the exact 68-symbol mapping is given in the official challenge inventory (IqraEval, 2026). This avoids a separate explicit CTC or forced-alignment decoding stage, though the model still learns implicit acoustic-text alignment through attention.

We used only the challenge data for adaptation: Iqra_train (~79 hours of error-free speech), Iqra_TTS (~80 hours with artificial mispronunciations), and Iqra_Extra_IS26 (~2 hours of real mispronunciations). We excluded 133 utterances (10%) of Iqra_Extra_IS26 from training. The remaining portion received a sampling weight of 20 and was combined and shuffled with Iqra_train and Iqra_TTS at the start of training to reduce data imbalance.

Configurations were selected using the official Dev split (2,588 Iqra_train utterances). The 133 excluded utterances were not used for checkpoint selection, hyperparameter tuning, early stopping, or other training decisions. After configurations were fixed, we evaluated an internal evaluation set of 433 utterances: these 133 real-error examples plus 300 randomly selected error-free Dev utterances. Thus, the real-error subset is independent of model selection, but the clean component overlaps Dev. This mixed set supports a matched-example comparison, not an independent test estimate, and its F1 is not directly comparable to hidden QuranMB.v2.

3.2 Qwen-MDD-AR

The system builds on Qwen3-ASR-1.7B (Shi et al., 2026). We froze the audio encoder and adapted the autoregressive decoder with LoRA (Hu et al., 2021), targeting q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj with rank $r = 64$, $\alpha = 128$, and dropout 0.05. We also unfroze embed_tokens and lm_head so the model could emit the new phoneme tokens. This left 700,551,168 trainable parameters, about 25.6% of the 2.04B logical parameters that include the audio encoder; the PEFT wrapper reported 2.738B total because it retained separate trainable copies of the token embeddings and language-model head.

The encoder was frozen to retain its pretrained acoustic representation and reduce adaptation memory; partial encoder fine-tuning was not evaluated. LoRA rank 64 and $\alpha = 128$ were practical capacity choices under the compute budget, not experimentally established optima. Encoder freezing, LoRA, phoneme-token extension, and 20-fold oversampling were not independently ablated.

Training used an effective batch size of 32, learning rate 5×10^{-5} , cosine scheduling with 0.05 warmup, maximum gradient norm 1.0, and maximum audio duration 100 seconds. The curriculum first fine-tuned without augmentation to 13,500 steps, then continued with SpecAugment (Park et al., 2019) to 19,000 steps, and finally used the stable 13,500-step checkpoint with SpecAugment plus 0.1 label smoothing to produce the 25,000-step submission. Training ran on a single NVIDIA L4 (24 GB); total time was approximately 40 h 24 m, with the final 19,000-to-25,000-step stage measured at 9 h 41 m 50 s. Peak GPU memory was not recorded.

3.3 CTC Baseline

For a matched-data comparison, we trained a Wav2Vec2-BERT CTC baseline (Communication et al., 2023; Graves et al., 2006) on the same oversampled challenge data. It was optimized for 26,000 steps with effective batch size 32, learning rate 1×10^{-5} , linear scheduling, 500 warmup steps, weight decay 0.01, and maximum gradient norm 1.0.

4 Results

4.1 Evaluation Metrics

Iqra’Eval ranks systems primarily by phoneme-level F1. For internal evaluation, we use the same evaluation implementation for both models: `align_data.py` aligns canonical, human-annotated, and predicted sequences using `metric.py`; `ins_del_cor_sub_analysis.py` computes detection and diagnosis metrics. True Accept (TA) and False Reject (FR) count correctly pronounced events accepted and rejected, respectively; True Reject (TR) and False Accept (FA) count actual pronunciation errors detected and missed. With $T = TA + FR + TR + FA$,

$$P = \frac{TR}{TR + FR}, \quad R = \frac{TR}{TR + FA},$$

$$F_1 = \frac{2PR}{P + R}, \quad DA = \frac{TA + TR}{T}.$$

Detection Accuracy (DA) measures detection decisions; Correct Diagnosis (CD) is the fraction of TR events whose predicted realization matches the human annotation. TA/FR rates divide by $TA + FR$; TR/FA rates divide by $TR + FA$.

Separately, against the human sequence, let C, S, I, D denote matches, substitutions, insertions, and deletions, and $N = C + S + D$. Phoneme Error Rate (PER) is $(S + I + D)/N$, Accuracy is $1 - \text{PER}$, and Correct Rate is C/N . Only Accuracy and PER penalize insertions; PER is not generally $1 - \text{DA}$. The alignment treats supplied symbols as tokens and uses `<eps>` for gaps. We retain the leaderboard’s reported values separately from internal sequence metrics. The internal set has only 122 canonical-human error events, making precision and F1 sensitive to false rejections; its F1 cannot be directly compared with hidden QuranMB.v2.

4.2 Baseline Comparison: CTC vs. Autoregressive LALM

To compare a CTC baseline and our frozen-encoder generative approach under matched data conditions, we evaluated both the Wav2Vec2-BERT baseline and Qwen-MDD-AR on the exact same subsets. The results of this controlled comparison are detailed in Table 2.

The Wav2Vec2-BERT baseline achieved lower PER on both Dev (0.0515 vs. 0.0649) and the internal evaluation set (0.0472 vs. 0.0541). It also achieved higher Accuracy (0.9528), precision (0.1608), and F1 (0.2761) on the internal set, making it stronger on aggregate alignment and detection metrics.

Qwen-MDD-AR matched Recall (0.9754) and obtained higher CD (0.9496 vs. 0.9328), corresponding to 6 vs. 8 diagnosis errors among 119 True Reject events. Because this is a two-error difference without a significance test, we treat it as preliminary. The identical FA rate (0.0246) is not a positive differentiator.

The comparison suggests a trade-off rather than a clean winner. The CTC model is better on alignment-sensitive aggregate metrics, as expected for a monotonic alignment objective. Qwen-MDD-AR is competitive despite a frozen encoder and generative decoder, but the internal evidence does not establish a diagnostic advantage.

4.3 Error Analysis

Against the human pronunciation on the 433 utterances, Qwen-MDD-AR makes 440 substitutions,

Table 1: Qwen-MDD-AR results on the hidden QuranMB.v2 Iqra’Eval 2026 leaderboard.

Team	F1	Precision	Recall	Correct Rate	TA	FR	FA	CD	PER
Organizer’s baseline	0.4414	0.3039	0.7707	0.8385	0.8763	0.1237	0.2293	0.612	0.1308
Qwen-MDD-AR (25k steps)	0.6702	0.6666	0.6738	0.9379	0.9758	0.0242	0.3262	0.8508	0.0445
Qwen-MDD-AR (19k steps)	0.6670	0.6678	0.6663	0.9378	0.9762	0.0238	0.3337	0.8462	0.0446
Qwen-MDD-AR (13.5k steps)	0.6288	0.5855	0.6791	0.9263	0.9654	0.0346	0.3209	0.8326	0.0538

Table 2: Matched-data comparison on Dev and internal evaluation sets. Internal PER is recomputed from the stored human–prediction alignments.

Metric	Wav2Vec2-BERT	Qwen-MDD-AR
Dev Set (2,588 samples)		
PER	0.0515	0.0649
Internal Evaluation Set (433 samples)		
PER	0.0472	0.0541
Detection Accuracy	0.9600	0.9573
Accuracy	0.9528	0.9459
Correct Rate	0.9596	0.9569
MD&D Standard Metrics		
F1-Score	0.2761	0.2633
Precision	0.1608	0.1522
Recall	0.9754	0.9754
MD&D Detection Rates		
True Accept (TA)	0.9599	0.9572
False Reject (FR)	0.0401	0.0428
True Reject (TR)	0.9754	0.9754
False Accept (FA)	0.0246	0.0246
Correct Diagnosis (CD)	0.9328	0.9496

171 insertions, and 232 deletions, versus 342, 106, and 288 for the CTC baseline (15,591 aligned phones; the internal PER values in Table 2 come from these alignments). The CTC model trades 98 fewer substitutions and 65 fewer insertions for 56 more deletions. Qwen-MDD-AR is exact on 213/433 utterances versus 196/433, but its errors are heavier when they occur: 90th-percentile utterance PER is 17.25% versus 15.31%, with 41 versus 28 utterances containing runs of ≥ 3 consecutive errors. A paired-utterance bootstrap puts the PER gap at 0.686 points (95% CI 0.239–1.149, $p \approx 0.002$).

This pattern is consistent with the decoder formulations. The autoregressive decoder conditions each token on its output history, so a wrong local hypothesis can cascade into coherent substitution/insertion bursts, while CTC’s frame-level blank/repeat collapse keeps errors acoustically local but omits weakly evidenced phones. Qwen’s output length is nearly unbiased (hypothesis/reference ratio 0.996 vs. 0.988), and errors are flat across utterance thirds, contradicting end-of-sequence drift. Vowel-quality confusions dominate for both models (58.6% of Qwen substitutions vs. 68.1% of CTC substitutions involve vowel-like symbols on both sides); Qwen additionally substitutes rather than deletes on emphatics and pharyngeals. Two utterances illustrate the complemen-

tarity: on audio_73881 Qwen-MDD-AR makes 2 substitutions, 2 insertions, and 2 deletions while the CTC baseline is exact, whereas on audio_5069_57 Qwen-MDD-AR is exact while the CTC baseline makes 3 substitutions and 1 deletion. With only 122 real error events, these observations are descriptive and exploratory; architecture is a plausible mechanism, not an isolated cause.

4.4 Official Challenge Results

On hidden QuranMB.v2, the 25,000-step checkpoint ranked 6th of 19 teams with 0.6702 F1 and 0.8508 CD (Table 1). The 13,500- and 19,000-step submissions achieved 0.6288 and 0.6670 F1, respectively. Leaderboard scores were available after each submission, but they were not used for tuning; configuration decisions were based on the official Dev split. Each stage was trained to convergence, evaluated, and then continued from the stable checkpoint with additional regularization. This is a sequential development history, not an ablation: training duration, SpecAugment, and label smoothing vary jointly. The final combined configuration obtained the best hidden-set F1 among our submissions, but these results do not isolate any regularizer’s effect. We used no additional task-specific external data or custom acoustic-pretraining stage; Qwen3-ASR itself is extensively pretrained.

5 Conclusion and Future Work

We introduced Qwen-MDD-AR, an autoregressive PEFT-based pipeline for Arabic MDD, and its first empirical investigation with a matched-data CTC baseline under challenge constraints. Our submission achieved 6th of 19 teams with an F1-score of 0.6702 using only provided Iqra’Eval 2026 data.

Our comparative analysis revealed a trade-off: the Wav2Vec2-BERT CTC baseline maintained an advantage in Phoneme Error Rate (PER), precision, F1-score, and overall accuracy on our internal evaluation set, while Qwen-MDD-AR matched its Recall and showed a small Correct Diagnosis difference (0.9496 vs. 0.9328). We treat this diagnosis result as preliminary because it corresponds to only

two fewer errors and has not been tested for statistical significance. Overall, Qwen-MDD-AR is best viewed as a promising LALM-based alternative for Arabic MDD rather than a replacement for strong CTC systems.

The main practical lesson is that LALM adaptation can be made simple: the system does not require a custom acoustic pre-training stage, and most of the backbone remains frozen. At the same time, the controlled baseline shows why simplicity alone is not enough. Any future LALM-based MDD system should be evaluated against strong CTC models on both alignment-sensitive metrics and diagnosis-specific metrics, and should report inference cost in addition to trainable-parameter count.

Future work should evaluate multi-stage domain adaptation and reference-guided prompting, while testing whether reference context improves diagnosis or masks real pronunciation errors. A reference-guided prompt is especially attractive for Quranic recitation, where the intended text is known in advance; however, this same prior can encourage the model to normalize away acoustic evidence. Error-type analysis over substitutions, insertions, and deletions would help determine whether the reference improves genuine diagnosis or merely improves fluent phoneme generation.

We plan to release the training and inference code, tokenizer modifications for the 68 phoneme tokens, configuration files, evaluation scripts, and the LoRA adapter, subject to the dataset license.

6 Limitations

The CD difference in the internal comparison is small (6 vs. 8 diagnosis errors among 119 True Reject events), and we do not report a significance test for it; only the recognition PER gap is bootstrap-tested. The internal evaluation set and hidden QuranMB.v2 have different error distributions, so their F1-scores should not be compared directly. We also lack detailed speaker demographics, phoneme-level error distributions, and linguistic-complexity statistics for the hidden test set, limiting deeper error analysis.

Our efficiency discussion is limited to training and adaptation: LoRA reduces trainable parameters relative to full SFT, but autoregressive inference with a 1.7B decoder is likely heavier than a CTC forward pass, and we do not report latency, FLOPs, or throughput. Finally, this challenge-system paper

does not compare against GOP variants, RNN-T MDD systems, or non-Arabic benchmarks such as L2-ARCTIC, and it does not quantify the risk that autoregressive decoding may use linguistic priors to mask or "correct" pronunciation errors.

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